Topic 1: Linear Functions

Form M

ANSWER KEY

Name _____ Date _____

> **Directions:** Read each question carefully. You may use a calculator and the formula sheet. For multiple choice, fill in the circle next to **one** answer. Take your time and show your work in the boxes.

Formula & Reminder Sheet (you may use this during the test)

Slope / rate of change $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{change in } y}{\text{change in } x}$	**Slope-intercept form** $y = mx + b$ m = slope, b = y-intercept
Point-slope form $y - y_1 = m(x - x_1)$ (x_1, y_1) is a point on the line	**Standard form** $Ax + By = C$ Solve for y to find the slope.
Intercepts x-intercept: where the line crosses the x-axis ($y = 0$) y-intercept: where the line crosses the y-axis ($x = 0$)	**Correlation coefficient r** Close to 1 or -1 → **strong** Close to 0 → **weak** $r > 0$ positive, $r < 0$ negative
Calculator: line of best fit STAT → EDIT: type x in L1, y in L2 STAT → CALC → LinReg(ax+b)	**Domain** = all possible x-values (inputs) **Range** = all possible y-values (outputs) **Causation**: one event <u>makes</u> the other happen

1 The table shows Lucia's salary. Let x = the number of years **since 2000**.

Year	2000	2001	2002	2003
x **(years since 2000)**	0	1	2	3
Salary ($)	40,000	42,000	46,000	44,500

Which linear function **best models** the data?

> Tip: Use LinReg on your calculator (see the formula sheet).

- ○ **A.** $y = 4375x + 39{,}625$
- ● **B.** $y = 1750x + 40{,}500$
- ○ **C.** $y = 40{,}500x + 1750$

2 Which situation shows **causation**?

> Causation means one event **makes** the other event happen.

- ○ **A.** When you study for a test, your classmate studies too.
- ○ **B.** When you carry an umbrella to school, it rains.
- ● **C.** When you don't brush your teeth, you get cavities.

3 Nia sells cookies at a constant price. The table shows her profit.

Number of Cookies Sold	Profit (dollars)
4	−1.50
5	1.25
6	4.00
7	6.75

What is the **rate of change**? Include units.

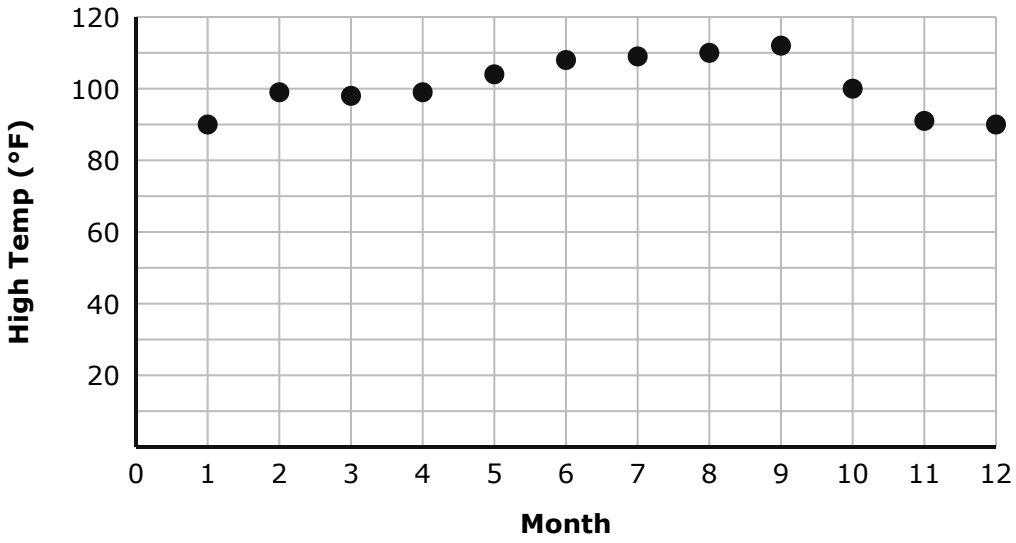
Step 1: Change in profit (from 4 to 5 cookies) = **1.25 − (−1.50) = 2.75** dollars

Step 2: Change in cookies = **1** cookie

Step 3: Rate of change = **2.75** dollars per cookie

4 The table and graph show the monthly high temperature in Austin, Texas.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Temp °F	90	99	98	99	104	108	109	110	112	100	91	90



What does the correlation coefficient say about the **strength** of the linear association?

> Find r with your calculator. Close to 0 = weak. Close to 1 or −1 = strong.

- ○ **A.** Weak negative correlation
- ● **B.** Weak positive correlation
- ○ **C.** Strong positive correlation

5 The table shows the pounds of chicken feed in a barn. Let x = months **since March**.

Month	Mar	Apr	May	Jun	Jul	Aug	Sep
x	0	1	2	3	4	5	6
Pounds (y)	950	873	796	719	642	565	488

Write a linear function for y, the pounds of chicken feed.

Starting amount when $x = 0$: $b =$ **950**

Change each month: $m =$ **−77**

$y =$ **−77** $x +$ **950**

6 The table shows how much Omar earns for the hours he works.

Hours (x)	0	1	2	3	4	5	6	7
Money, \$ (y)	0	16	35	47	60	69	90	103

What is the best prediction of what Omar earns for **9 hours**?

> Step 1: Use LinReg to find the line of best fit. Step 2: Put $x = 9$ into the equation.

- ○ **A.** \$128.00
- ● **B.** \$131.46
- ○ **C.** \$189.00

7 What is the **slope** of the line $15x + 3y = 45$?

> Solve for y to get $y = mx + b$.

Step 1: Subtract $15x$ from both sides: $3y =$ **−15x + 45**

Step 2: Divide every term by 3: $y =$ **−5x + 15**

Slope $m =$ **−5**

8 Write the equation in **slope-intercept form** of the line with slope $\frac{1}{3}$ through the point $(-1, -2)$.

> Use point-slope form: $y - y_1 = m(x - x_1)$. Here $m = \frac{1}{3}$, $x_1 = -1$, $y_1 = -2$.

Step 1: Fill in: $y -$ (**−2**) = **1/3** $(x -$ (**−1**))

Step 2: Simplify: $y + 2 = \frac{1}{3}(x + 1)$

Step 3: Distribute and subtract 2: $y =$

(1/3)x + 1/3 − 2 = (1/3)x − 5/3

Show your work

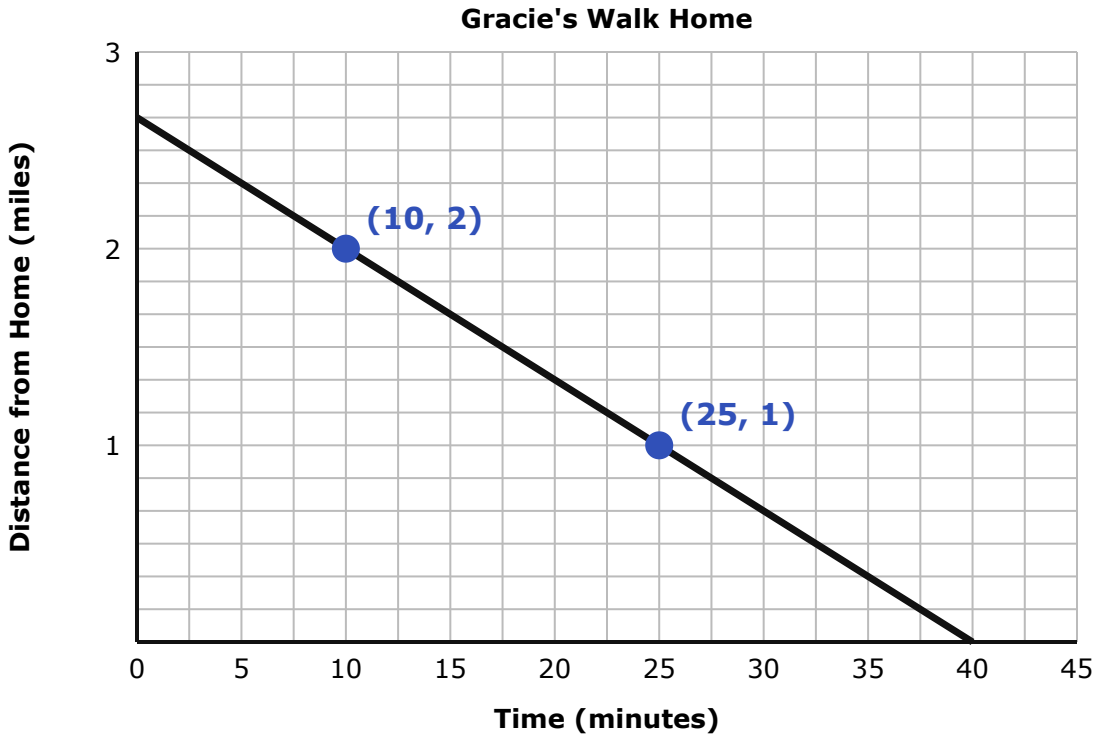
9 The water in a tub drains at a constant rate.

Time, minutes (x)	Water left, liters (y)
0	360
2	240
3.5	150
5	60

Step 1: Use two points, (0, 360) and (2, 240). Rate = $\frac{240 - 360}{2 - 0}$ = **−60** liters per minute

Step 2: Complete the sentence: The water level goes **up** / **down** (circle one) by **60** liters every minute.

Use the graph for questions 10, 11 and 12. The graph shows Gracie's walk home from school.



Each small box going up is **1/6 of a mile**. Each small box going across is **2.5 minutes**. Two points on the line are marked for you.

10 What is the x**-intercept**? Explain what it means.

x-intercept = (**40** , 0)

Meaning: After **40** minutes, Gracie is **0** miles from home. (She is **home** .)

11 What is the y**-intercept**? Explain what it means.

y-intercept = (0, **2 2/3**)

Meaning: At 0 minutes (when she leaves school), Gracie is **2 2/3** miles from home.

12 Find the **slope**. Explain what it means.

Use the marked points (10, 2) and (25, 1): $m = \frac{1 - 2}{25 - 10} =$

−1/15 mile per minute

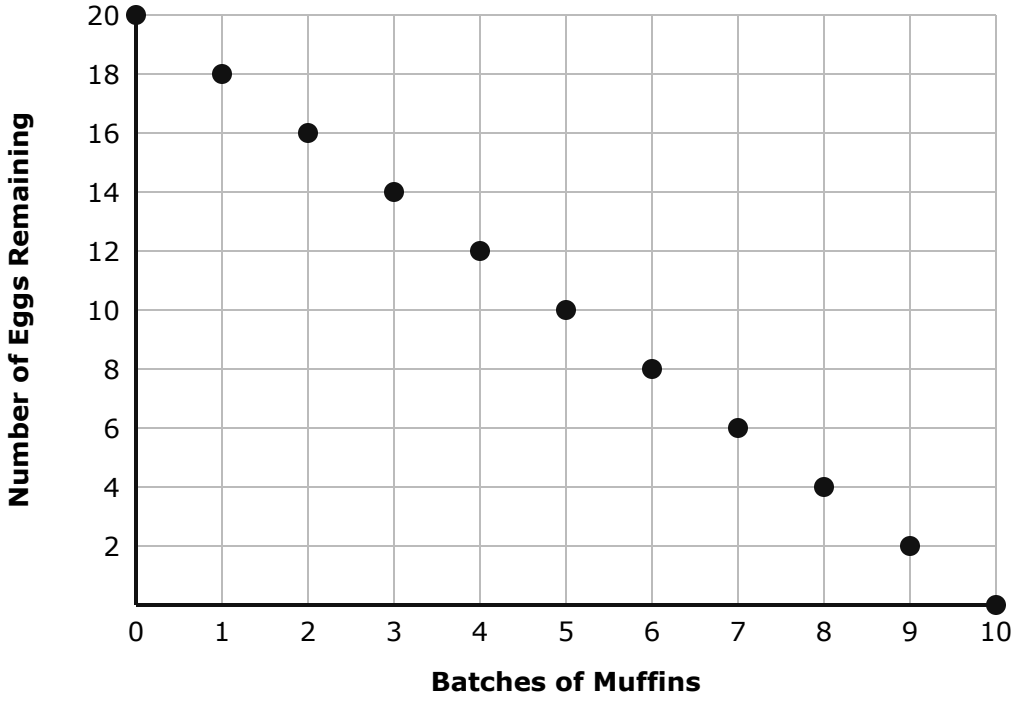
Meaning: Gracie gets **1/15** mile **closer to** / **farther from** (circle one) home every minute.

13 What is the value of $f(x) = 23.2x + 2.5$ when $x = -4.2$?

> Replace x with -4.2: $f(-4.2) = 23.2(-4.2) + 2.5$. You may use a calculator.

- ○ **A.** −99.94
- ● **B.** −94.94
- ○ **C.** 21.5

14 Miguel has 20 eggs. Each batch of muffins uses 2 eggs. The graph shows y, the eggs left, and x, the batches he bakes.



What is the **domain** of the function?

> Domain = the x-values. Look at the points on the graph. He can only bake **whole** batches.

- ○ **A.** $0 \leq x \leq 10$
- ● **B.** {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
- ○ **C.** {0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20}